
NOVEL FEATURES EXTRACTION PIPELINE FOR SKIN LESION IMAGES

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ABSTRACT

According to the *International Agency for Research on Cancer (IARC)*¹, solely in 2022, 1.5 millions of skin cancer cases have been documented, and of those, 60 thousands died for malignant melanoma. To be able of reducing these statistics, several Artificial Intelligence based solutions have been tested, with decades of literature behind.

Lately, most of the research groups effort goes in building novels deep learning architectures able to accurately classify a skin lesion from a unique picture, reaching high accuracy levels. Unfortunately these architectures lack in interpretability, thus leading field specialists to beware of them. To make up for this flaw, this project focuses on building a novel pipeline that processes noisy skin lesion images to compute discriminative hand-crafted features based on the clinical *ABCD rule*², such asymmetry, border irregularity and color variation, that are interpretable for a specialist.

Before being able to extract those features, one image is first resized to a standard size, then body artifacts, such body hairs, are removed and the image reconstructed, consequently the lesion is isolated from the skin background with the aid of segmentation techniques. Once the pre-processing is complete, features are extracted from the dataset and used to train machine learning classifiers. Each step of the proposed pipeline has been individually evaluated, to assure optima performances.

The final pipeline has been used to process the HAM10000 dataset, preparing it to train a Random Forest classifier, which aim is facilitate the first screening of a lesion. The classifier trained to classify a lesion in "Suspicious" and "Non suspicious" obtained a F1-score of 70%, with a specificity and sensibility of 70% on average, as well. Along with the classifier, the SHAP framework has been used to compute the importance that each feature had in making the prediction, allowing to evaluate if the model decision is based over strong and correct assumptions.

1 Introduction

According to *GLOBOCAN 2022*³, in the world there are:

- over 1.5 millions of new skin cancer cases.
- over 330 thousands of new Melanoma cases.

¹Specialized cancer agency of the World Health Organization (WHO)

²Asymmetry, Border, Color, and Diameter. These are guidelines proposed by dermatological councils to assess the risk of malignant skin tumors. [<https://www.aad.org/public/diseases/skin-cancer/find/at-risk/abcde>]

³IARC project that records and distributes statistics regarding the main kinds of cancer in a total of 185 world countries.

- over 60 thousand deceased for Melanoma.

The best strategy to minimize these numbers is a fast and well-formulated diagnosis of the patient conditions, unfortunately not always available due to numerous reasons. As stated by *BMJ Quality & Safety* article: "variations in gps' decisions to investigate suspected lung cancer: A factorial experiment using multimedia vignettes" [1], the main issues in a correct and timely diagnosis of a lung cancer, that can be generalized to any diagnosis approach, are:

- Lengthy diagnostic process.
- A misjudgment by the doctor of their own abilities, due to either excessive self-confidence or, conversely, low self-esteem.
- Patient personal characteristics.

To minimize these criticality, the use of *Artificial Intelligence (AI)* tools has been widely adopted, leading to exceptional results in numerous fields.

In the dermatological domain, several approaches to automatically detect the presence and type of skin tumor given a patient picture have been developed and tested, reaching high level of accuracy in detecting the disease, but at the cost of losing notion on how these intelligent systems reached their conclusion.

These trade off between efficiently detecting the disease and working with an interpretable and explainable approach exists because the main methodology of *Computer Vision* aided system in dermatological classification of skin lesion employees *Convolutional Neural Network (CNN)* or *Vision Transformer (ViT)*, highly efficient in finding underlying patterns in pictures that are useful to correctly classify them, but that lacks in *interpretability*⁴

To overcome the issues brought in by these deep neural architectures several approaches can be implemented. These range from evaluating what parts of the image is determinant for the CNN decision, possible with algorithms like *Grad-CAM*[2], to bypass these architecture entirely. In the latter, hand-crafted features can be manually extracted from the images to train *Machine Learning (ML)* models that are interpretable for a specialist, and their reasoning process can be used to effectively accelerate the diagnostic process by providing that can give means to contradict or confirm the model outcome, resulting in a more reliable second opinion.

The proposed pipeline of actions takes in one image, resize it in a standard size of (256, 256), and removes accentuated body artifact from it. After these firsts steps, the skin lesion is successfully separated from the image background with segmentation techniques and then trans-rotated according to its axes. With the segmented, clean image, the chosen hand-crafted features resembling the ABCD rules can be extracted. For evaluating asymmetry, two independent indexes are used, both in a continuous range of [0; 1], one measures the shape asymmetry, the second one the color asymmetry. The border irregularity is assessed with an index in a discrete range of (0, 8). Next, the presence of the main structures that works as guideline during the diagnostic process is identified by extracting key features from the image and train classifiers to detect their presence. The structures to be identified are:

- Pigment Network;
- Streaks;
- Regression Areas;
- Blue-Whitish Veil.

Initially, also the presence of *Globules* structures was considered to be an input feature, but since the poor performances obtained in extracting this kind of features from the images, it would have ended up to be only added noise.

By combining these features extractors, the pipeline is complete and ready to be used.

The extracted features are used to train a Random Forest classifier, enriched with a *SHAP* framework⁵, that provides indications on which features combination is determinant when providing an explanation.

⁴Degree to which machine learning algorithms can be understood by humans. More specifically, interpretability describes the ability to understand the reasoning behind predictions and decisions made by a machine learning model. [<https://it.mathworks.com/>]

⁵SHapley Additive exPlanations

2 Related Work

To build the pipeline, numerous studies were consulted, looking for tested strategies that have already brought in verified results. Firstly, to overcome the issue of having image noise, the images of the used datasets must be properly de-noised, by removing hair artifacts. The main article used as guideline is "*Melanoma skin cancer detection method based on adaptive principal curvature, color normalization and feature extraction with the abcd rule.*", published on *Journal of Digital Imaging*"[3]. The authors used an Hessian matrix with a gaussian kernel to compute the principal curvatures of each pixel, by assuming that one hair follows a Gaussian distribution, artifacts that matched the filter should be detected. The method proposed in this project instead of explicitly doing the matching, uses the *Frangi filter*[4], usually used to detect tubular structures. The filter is processed in the value channel of the *HSV color space*⁶, since similar works points out improvements when operating within this channel[5]

To segment the lesion, the state of the art relies heavily on U-Net architectures[6], and although newer and more performing architectures based on attention mechanisms have been developed, the results offered by a simple U-Net architecture, when trained with properly cleaned images, were highly satisfactory.

To determine which features should be extracted and to define the corresponding methodologies, the foundational concepts were derived from both "*Implementation of the 7-Point Checklist for Melanoma Detection on Smart Handheld Devices*" works, discussed during the *Conference Proceedings of the IEEE Engineering in Medicine and Biology Society (EMBC)* [7] and the *PH2 Database*, containing lesions from which hand-crafted features have been extracted by specialists[8].

By using the *PH2* features as guidance, several strategies have been implemented, and to evaluate the presence of each geometric structure robustly, specific algorithmic modifications were introduced.

3 Materials

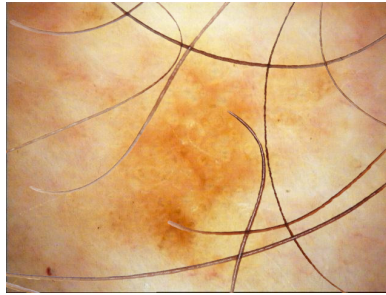
To appropriately build the final pipeline, several dataset has been used, since for each tasks a precise set of data is needed.

The Skin Hair Dataset (SHD)[9], is made of 252 images originally from the *ISIC archive*⁷, has been deployed to assure a common benchmark for the hair removal task.

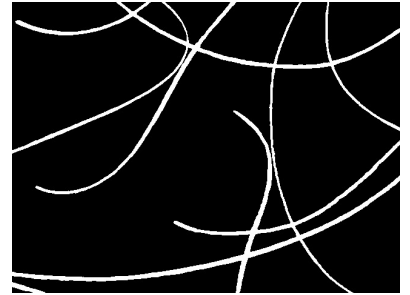
For each photo an augmented one with hand-made hair and relative mask included.



(a) Picture 1:Original



(b) Picture 1:Augmented



(c) Picture 1:Mask

The *Human Against Machine with 10015 training images (HAM10000)*[10], is one of the most famous open-source benchmark for skin lesion classification, providing for each image a mask of the lesion, final diagnosis and meta-data such age, sex and body location of the lesion on the patient. It has been used to train a U-Net for segmenting skin lesions, since the dataset provides the masks, and afterwards adapted with the final pipeline to train the classifiers. The dataset distribution over the diagnosis can be seen in the graph 2

Ultimately, although the features extraction pipeline has been modeled with three different classes, Melanocytic Nevus, Melanoma and Actinic Keratoses, the remaining classes of HAM10000 have been merged, allowing a classification for the Common Nevus, the Malignant lesion, primarily melanoma, and the precancerous form, the actinic keratoses. The final distribution can be seen in the graph 3.

⁶Color space in which colors are represented based on how humans naturally perceive them, using three components: Hue (the color family), Saturation (the purity or intensity), and Value (the brightness or darkness).

⁷The International Skin Imaging Collaboration (ISIC) Archive is the largest public repository of dermatological images, established to support global research in automated skin cancer diagnosis and computer vision.

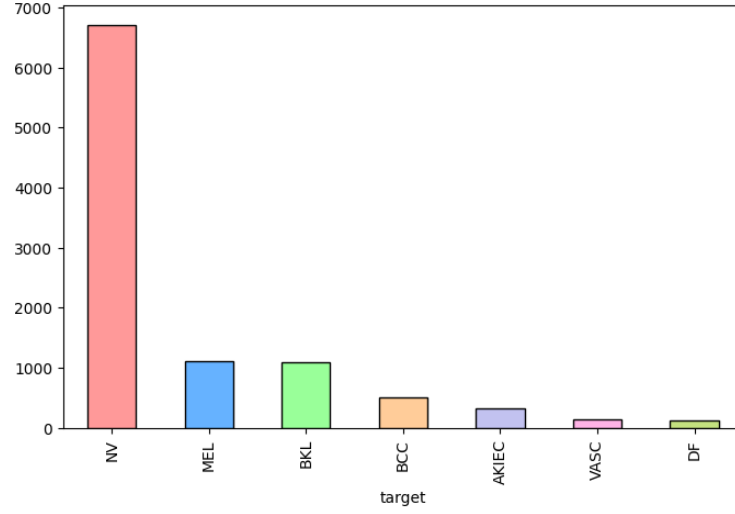


Figure 2: HAM10000 target distribution

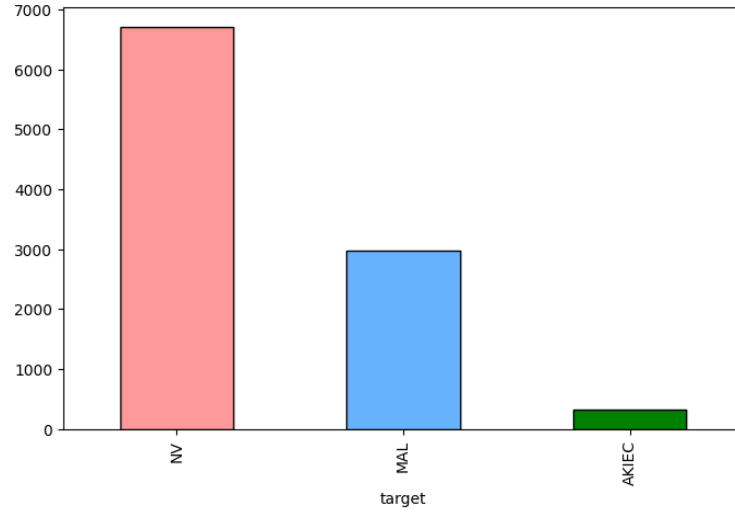


Figure 3: Dataset final distribution

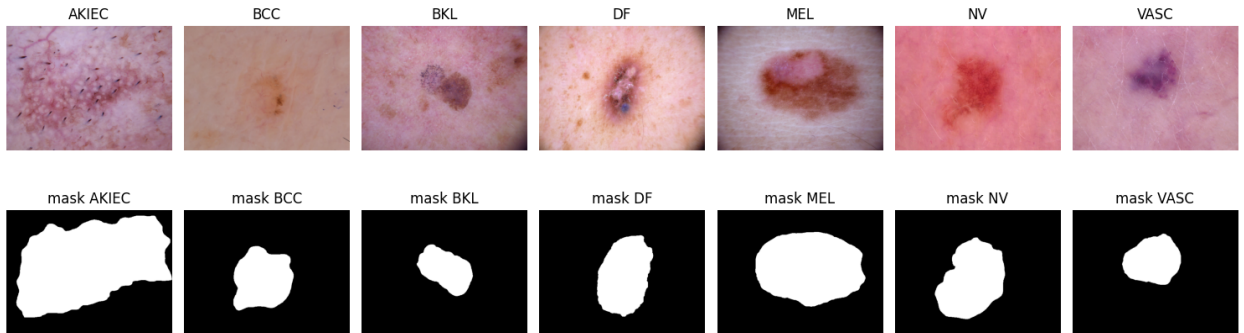


Figure 4: HAM10000 class samples.

Finally, the data used to build and validate the final pipeline come from the *PH2 dataset*[8], result of the collaboration between the *University of Porto*, the *Higher Technical Institute of Lisboa* and from the *Hospital of Pedro Hispano* of

Matosinhos, Portugal.

The dataset counts 200 images distributed in three classes, Common Nevus, Atypical Nevus and Melanoma, as can be seen in 5. For each lesion a mask of the lesion is shared.

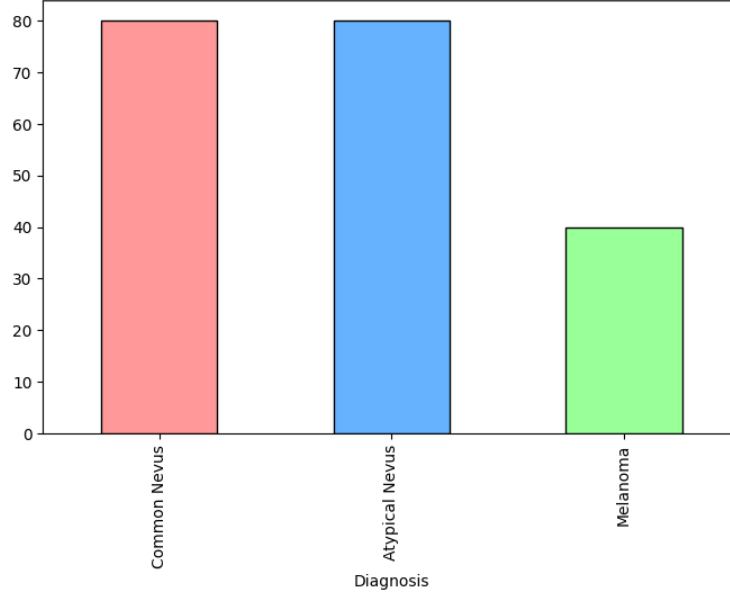


Figure 5: PH2 target distribution

4 Methods

The following section focuses on the implementation choice made during the design of each one of the modules that compose the final pipeline, as well as the model chose for the final classification.

All images has been resized to a (256, 256) shape before training each one of the modules pipeline, 6.

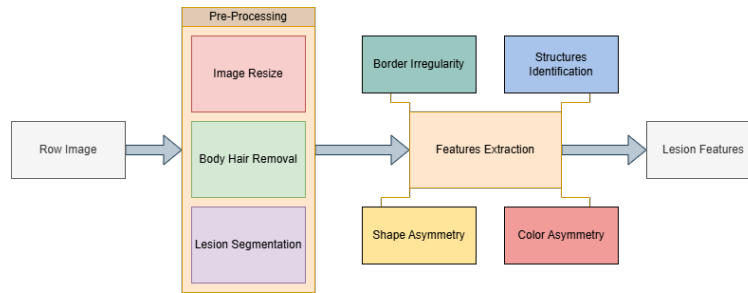


Figure 6: Final Pipeline

4.1 Features Extraction Pipeline

The first module implemented is the "*Body Hair Removal*".

To accurately segment the body hair from the image, the image has been first converted in the *HSV color space*, then the Frangi filter has been applied over it. The resulting mask has been normalized with the normalization function in the *OpenCV2* library, then slightly blurred with a gaussian filter to reduce the noise made during the segmentation, enclosing possible white-dots inside body hair shape. The use of second filter to enclose open shape, based over the Haralick circularity index, has been evaluated but ultimately avoided due poor performances.

To make up for the parts of the image were before there were body hairs, now empty, the in-paint function of OpenCV with *TELEA* strategy has been used.



Figure 7: Example of image cleaned from body hair artifacts.

The second module is the "*Lesion Segmentation*".

Two different strategies have been evaluated. First, using image operation to enhance the difference between the skin lesion and its background, accurately extracting it. The latter consisted in training a U-Net classifier over the HAM dataset, learning to distinguish a lesion from its background.

The first attempt first sharpens the image with a filter, then uses the *Otsu Thresholding* to compute a raw mask. The mask is processed, maintaining the wider part of the area detected, later morphological operations are used to eventually fill gaps that may be present in the detected lesion.

The U-Net architecture has been taken from the *segmentation models pytorch* library, and evaluated with different backbones, *resnet50* and *efficientnet*, different image color space, and by varying the number of epochs, learning rate and batch size. The result of the network can be seen in picture 10

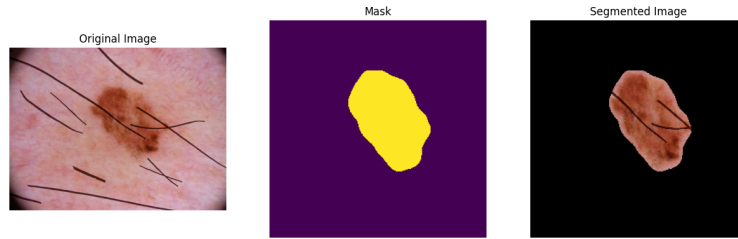


Figure 8: Example of segmented lesion.

The third module is the "*Feature Extraction*".

The lesion is first translated and rotated using a trans-rotational matrix, that rotates the image according to its barycenter, using its spatial moments.

To assess the asymmetry level of the lesion, two asymmetry indexes are computed, one that evaluates in a continuous range $[0, 1]$ the shape asymmetry of the lesion, the other that evaluates the color asymmetry of the lesion, also in a range $[0, 1]$.

The shape asymmetry is computed by splitting the image mask in its barycenter, and for each axes, computing a *bit-wise xor* between the two parts to check the area that doesn't overlap, the result is split by two and normalized for the total area of the lesion.

For the color asymmetry a similar methodology is applied, but an image in the LAB color space is used, since the euclidean distance between two points in this color space resembles the difference perceived by the human eye. The lesion is mirrored over the two axes and the distance between a pixel and its mirrored partner is computed, only for those pixels that have a mirrored partner, otherwise a lesion pixel could be confronted with a background one. The distances between each axes are averaged, and the two index are merged with the euclidean distance, then normalized using a sigmoid function that squash the index in a value between 0, colors equally distributed, and 1, maximum heterogeneity between color in lesion areas.

$$normalized_asym = \frac{2}{1 + e^{-k \cdot x}} - 1, k = 0.5$$

To assess the border irregularity of the lesion, the implemented strategy follows the algorithm first presented in "Novel Method for Border Irregularity Assessment in Dermoscopic Color Images"[11].

As the paper explains, the contour of the lesion is extracted, and for each point of it, the distance from it and the barycenter of the lesion is computed. Right after, a signal is designed over these distances, and is split in 8 octants.

For each octant, a gaussian signal, modeled after the extracted one is built and subtracted, allowing to emphasize the differences between the original border and what a regular border should be like, as can be seen in figure 9. To avoid that noisy irregularities bring distorted results, a threshold for the noise is implemented.

If one octant has at least one irregularity, a point to the score is added, for a maximum of 8 and a minimum of 0. By adopting this strategy, wider lesion won't be considered more irregular than tinier ones.

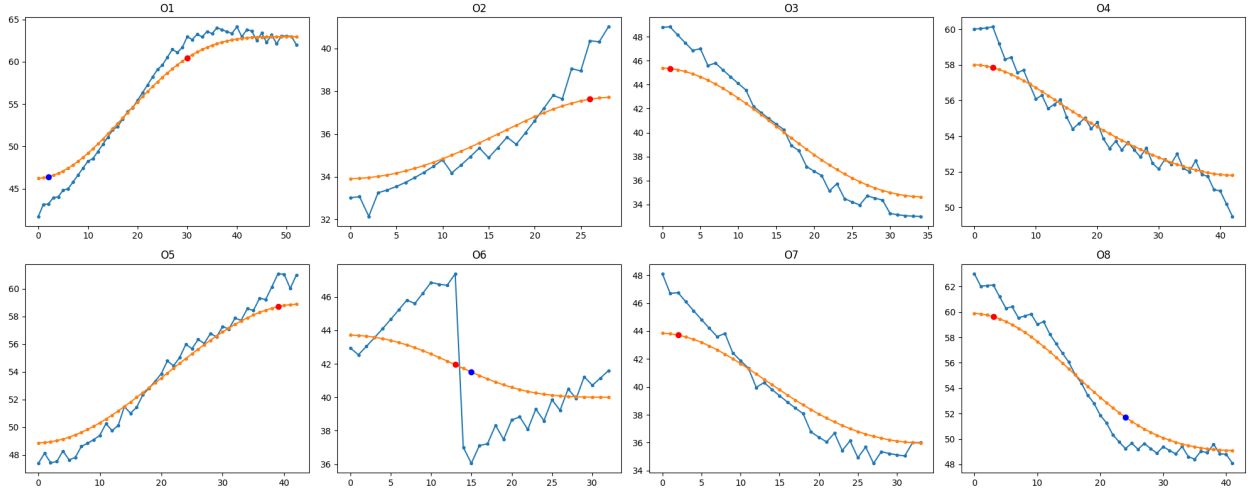


Figure 9: Example of octant irregularities. The blue signal represent the lesion border, while the orange one is a gaussian signal modeled over the border one.

Lastly, the presence of key geometrical features has been assessed by using machine learning models.

Images were first preprocessed, by extracting the key features described in the seven points paper, but adding also the Haralick features⁸ [7]:

- Key structure patters. The image is split in patches, for each one the Haar wavelet function is computed and mean and variance collected, allowing for a k-means model, previously trained, to classify each patch to a cluster that represents a kind of pattern; finally, the number of individuated pattern is collected.
- Local Binary Patterns (LBP)⁹. From the image the LBP are collected, using the LBP_{16} and LBP_{24} following a rotation invariant method; the collected patterns are used to build two different histograms.
- The image is then converted in two different spaces, i. e. HSV, CIELAB¹⁰, not including the RGB color space as proposed by the original paper since the poor information it would have added when dealing with non standard illuminated images. For each of these channels histogram, standard deviation and chromatic entropy are collected.

⁸statistical features that are computed over the entire image, utilized to describe the overall texture of the image using measures such entropy and sum of variance

⁹highly efficient visual descriptor widely used in computer vision for texture classification. It operates by comparing the grayscale value of a central pixel with its surrounding neighborhood; if a neighboring pixel's intensity is greater than or equal to the center, it is assigned a 1, otherwise a 0. These values are then concatenated into a binary number, effectively capturing local spatial structures such as edges, corners, and spots.

¹⁰

- Lastly, the Haralik features collected from the "Gray-Level Co-occurrence Matrix"¹¹. The features are the Angular Second Moment, or energy¹², Contrast¹³, Entropy¹⁴ and lastly the Inverse Difference Moment¹⁵.

All these features are combined in a vector, ready to be used as input for the classifiers.

To be able of considering such structures, Support Vector Machines (SVM) were designed as proposed by the paper. For each geometric structure, an individual SVM have been developed, using a One-vs-One strategy for the SVM trained to classify the three different classes of the Globulin Structure.

Unfortunately, since the poor results of the designed classifier in classify the globulin structure, it was not considered in the final dataset.

For each model, the hyperparameters where searched with a complete Gridsearch strategy focused on increasing the 'balanced accuracy' that is a mean between sensitivity¹⁶ and specificity¹⁷, since it represents a more reliable metric with respect to accuracy when dealing with unbalanced datasets, and focuses on reducing the number of false negative, fundamental when working with clinical purpose model.

The hyperparameters considered were:

- *Reducer* $\in (PCA, SelectKBest)$; the reducers works to reduce the number of features before giving them to the SVM. While the PCA projects the features in a new vector space with lower dimensionality but with maximized features variance, preserving a given percentage of information, the "SelectKBest" maintains only the K features that are statistically more relevant to the output.
- *Classifier Kernel* $\in (rbf, linear, polinomial)$; the standard SVM splits data with a linear boundary, thus with non-linearly separable data it would return poor performances. To avoid this inconvenience, the different kernel can be used to project data in a space where they could be separable. While the radial basis function, "rbf", uses a circle boundary, the polynomial relies on polynomial curves.
- $C \in (1e-1, 1e0, 1e1)$; regularization parameter that controls the compromise between the split boundary and the classification error. Higher C means wider boundaries margin, thus a more general model that accepts some errors.
- $\Gamma \in ('scale', 1e-1, 1e-2)$; regularization parameter used with rbf kernel, that defines how wide the influence of a point is. Higher gamma means tight local boundaries, thus more complex borders.
- $Degree \in (2, 3)$. Grade of the polynomial function used with the SVM polynomial kernel.
- *Class weights* : $(None, 'Balanced')$; regularization parameter necessary when dealing with unbalanced data, that increase the weight of misclassified minority classes samples.

The table 1 summarize the results obtained, excepts for the class weights hyperparameter, since for all the SVMs the best performances came from having non balanced weights.

Table 1: Best Hyperparameters for Support Vector Machines

Geometrical Structure	Reducer	Kernel	C	Gamma	Degree
Streaks	PCA (0.85 <i>inf</i>)	linear	0.1	None	None
Regression Areas	PCA (0.85 <i>inf</i>)	polynomial	1	None	3
Blue-Whitish Veil	SelectKBest($K = 30$)	polynomial	10	None	3
Pigment Network	SelectKBest($K = 100$)	rbf	0.1	0.1	None
Dots/Globules	SelectKBest($K = 50$)	rbf	10	0.01	None

¹¹Matrix that describes how gray pixel are disposed in the image with respect to the other pixels.

¹²Feature that measures the homogeneity of the image.

¹³Feature that quantify the local pixel variations.

¹⁴Feature that address the causality of the image, its heterogeneity.

¹⁵Feature that quantify how much close pixels are similar.

¹⁶True positive rate, that is the percentage of true positive correctly classified by the model

¹⁷True Negative Rate, that is the percentage of true negative correctly identified by the model

4.2 Classification Models

The classification model considered for the task is the Random Forest. To train it, the HAM10000 dataset has been processed with the developed pipeline, split in a 60% – 40% train-test stratified split, and its hyperparameter were tuned with a complete Gridsearch, with the following hyperparameters:

- *Estimators* $\in (50, 100, 500)$; that is the number of Decision Tree the model is made of.
- *Max Depth* $\in (None, 10, 20)$; it represents a boundary for the classifier, stopping it from going too deep with the splits and thus avoiding overfitting.
- *Min Samples Leaf* $\in (1, 5, 10)$; it defines the minimum number of samples necessary to perform a leaf. Increasing it means avoiding too specific leaves.
- *Max features* $\in (None, Sqrt, log_2)$; this parameter defines the maximum number of features that can be considered when splitting. If *sqrt* or *log₂* it is the square root, or logarithm, of the total features in input, otherwise all features.
- *criterion* $\in ('gini', 'entropy')$: it is the metric that measures the quality of a split. Gini refers to the Gini Impurity, calculated as $1 - \sum_{i=1}^n p_i^2$, where p_i is the probability of an element belonging to class i . It measures how often a randomly selected sample from the node would be mislabeled if it were assigned a label based on the class probability distribution in the node. Entropy, calculated as $-\sum_{i=1}^n p_i \log_2(p_i)$, measures the impurity or uncertainty of a node. When using entropy, the algorithm tries to maximize the Information Gain, meaning it seeks to reduce the dis-homogeneity and uncertainty of the resulting child nodes after the split.
- *balanced* $\in (balanced\ weight, RandomUnderSampler, None)$: since the class distribution is heavily unbalanced, the classifier has been trained by trying several adjustment to the data distribution. The classifier has been tested with no adjustment to class distribution, by decreasing the samples of each class until the distribution was equal for each target and by weighting the error for misclassifications based on the class distribution, so that wrong prediction over minority classes weight more than majority classes wrong prediction.

Furthermore, since the classifier aim is to help in making a preliminary diagnosis for the specialist aid, a second classifier has been trained in classifying a lesion in "Suspicious" and "Non-Suspicious" as well, providing a first screening. The same hyperparameters tuning done for the first classifier has been conducted for the binary one.

At last, the effectiveness of each features in making the prediction has been evaluated by using SHAP, allowing to overcome the "black-box" nature of these classifiers, and accompany the adopted classifier with an explanation behind its predictions.

The best hyperparameters find by Gridsearch are visible in the next table, 2:

Table 2: Best Hyperparameters for Random Forest

Classifier	Balance Strategy	Estimators	Max Depth	Min Samples Leaf	Max Features	Criterion
Multi-class	UnderSampling	500	None	5	None	Entropy
Binary Class	UnderSampling	100	10	20	sqrt	Entropy

5 Results

5.1 Body Artifact Removal and Lesion Segmentation

The techniques applied to remove body artifact returned remarkable results for a processing method that doesn't employ any complex model. The metric used to evaluate the inpainting is a metric that combine Peak Signal-to-Noise Ratio (PSNR)¹⁸ with the GLCM, evaluating both the color distribution and the local pattern preserved after the inpainting. The final score is 79%.

¹⁸standard metric used to evaluate the quality of image reconstruction or compression. It represents the ratio between the maximum possible power of a signal (e.g., 255 for 8-bit images) and the power of the corrupting noise, typically measured using the Mean Squared Error (MSE). Expressed in decibels (dB), a higher PSNR value indicates greater fidelity to the original image.

Speaking of lesion segmentation, the results are impressive. The metrics used to evaluate the U-Net are the Jaccard Index (IoU)¹⁹ and Dice Coefficient (DICE)²⁰, reaching an average of 90% for IoU and 95% for DICE.

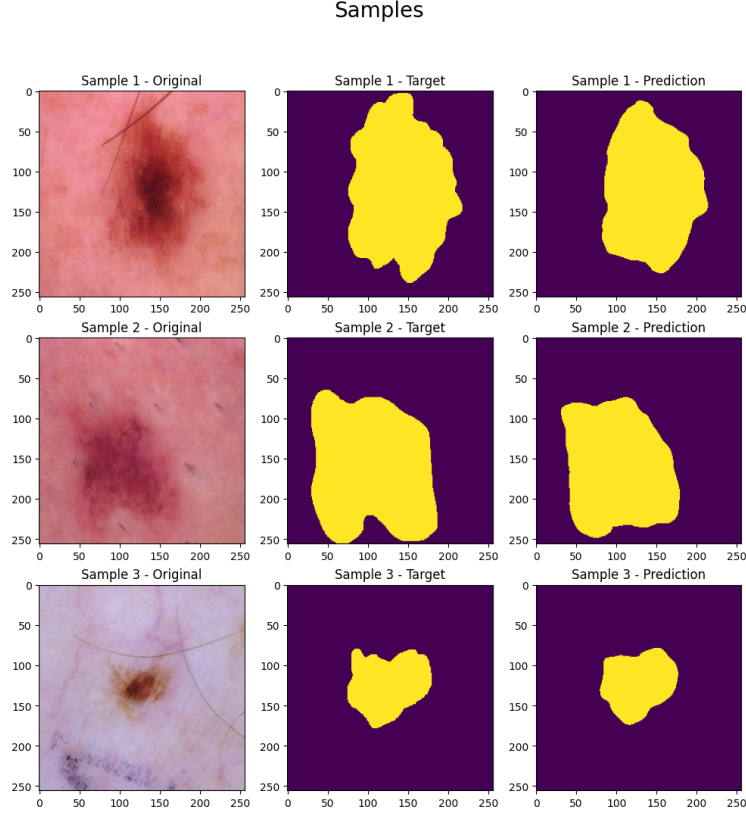


Figure 10: Example of segmented masks.

5.2 Geometrical structures

As can be seen in table 3, the classifiers demonstrate the most robust performances when detecting Blue-Whitish Veil and Streaks, both high reliable at distinguishing positive and negative cases.

Meanwhile, the Regression Areas classifier shows acceptable and balanced performances, with an optimum sensitivity, and a good specificity. Unfortunately, the Pigment Network classifier obtains poor performances, highlighting a general difficulty for the model in identifying the structure; as for the Pigment Network, the Dots/Globules classifier obtains even worst performances in identifying such structures, reaching an incredible high specificity, but a terrible sensitivity, meaning that most of the time the classifier doesn't recognize true positive cases.

5.3 Final Diagnosis

The Random Forest trained to classify three classes, Pre-Cancerous, Malignant and Benignant, shows difficulties in dealing with malignant and pre-cancerous classes, reaching a macro F1-score of 0.56, heavily influenced by the poor results in distinguish pre-cancerous and malignant lesions, as can be seen in table 4. The results show that, with only the extracted features, the model is unable to distinguish between malignant and pre-cancerous, highlight the difficulties of recognize complex information as the difference between these two classes.

This conclusion is supported by the results obtained with the binary classifier, that shows respectable performances in distinguish between Suspicious and non lesions, reaching a F1-score of 0.70, and balanced sensitivity and specificity, as can be seen in table 5.

¹⁹Ratio of area between predicted mask and provided one over the total area of both. $IoU = \frac{|P \cap G|}{|P \cup G|}$.

²⁰Considered as F1-score at pixel level, it is less strict then IoU. $Dice = \frac{2|P \cap G|}{|P| + |G|}$.

Table 3: Geometrical Structures classifiers results

Geometrical Structure	F1-score macro	Specificity	Sensitivity
Streaks	0.79	0.85	1
Regression Areas	0.66	0.77	0.80
Blue-Whitish Veil	0.84	0.94	0.82
Pigment Network	0.65	0.66	0.63
Dots/Globules	0.45	0.96	0.30

Table 4: Geometrical Structures classifiers results

Class	F1-score macro	Specificity	Sensitivity
Pre-cancerous	0.17	0.79	0.64
Malignant	0.40	0.77	0.38
Benignant	0.72	0.75	0.62
Average	0.43	0.76	0.55

Then, by using SHAP, the importance of each features in making the final classification has been measured, and the results for both model can be viewed in the figure ??.

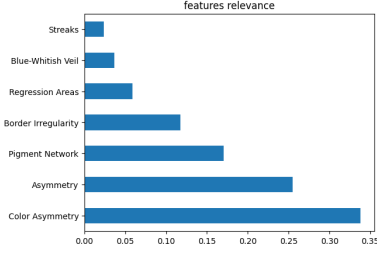
By comparison, can be confirmed the importance of Color Asymmetry in making a prediction, while Streak Structures play a marginal role. Is important to emphasize that the Streak Classifier is between the best classifiers found, meaning that the poor added value brought in the prediction can't be appointed to a bad classification in the pipeline.

Meanwhile, the main difference that can be notice is with the Blue-Whitish Veil. While, with the multi-class classifier it is an unnecessary feature, that plays a trivial role in the prediction, with the binary classifier is the second most important. As suspected, when facing the problem of classifying between suspicious and non, the Blue-Whitish Veil become a strong evidence, but if a further classification between pre-cancerous, benignant and malignant lesion has to be done, the presence of the veil doesn't bring in any new information. It is most likely this the reason behind the poor importance it has in the multi-class classifier.

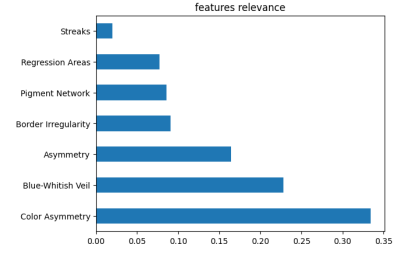
Another difference is between Asymmetry and Pigment Network, that played an important role in the multi-class classifier, likely to overcome the problem of distinguish between benignant and pre-cancerous, diversity that couldn't be denoted by the whitish veil presence.

Table 5: Geometrical Structures classifiers results

Class	F1-score macro	Specificity	Sensitivity
Suspicious	0.62	0.71	0.72
Non-Suspicious	0.77	0.72	0.72
Average	0.70	0.72	0.72



(a) Picture 1: SHAP over Multi-Class Random Forest.



(b) Picture 1: SHAP over Binary Random Forest.

6 Conclusion

The results allow us to confirm the validity of the pipeline for first level screening, since it shows good and stable performances in extracting features for distinguish between a suspicious lesion and a likely benignant one.

Unfortunately, it can't be used for a more advanced test, since the multi-class model trained with the extracted features couldn't effectively distinguish between the three classes, probably because of the poor performances in distinguish fine structures as Pigment Network and Globules. But the use of SHAP allowed us to confirm the validity behind the binary classifier, which uses features as color asymmetry and Blue-Whitish veil as key factor in making a prediction, as suggested in clinical research, and emphasize the poor value added by Streaks structures.

The project next steps should be focused in improving the classifiers behind Dots/Globules and Pigment Network structures, bringing new useful information to the pipeline and improving the performance in more complex screening. Focusing on detecting the range of colors that compose the lesion could be an added value to the pipeline, improving the information brought in by the color asymmetry, that already proved itself as one of the key features in making a prediction.

Finally, an application that incorporates pipeline and classifier, as well as other post-hoc evaluation technique as LIME²¹, could be built and tested with specialist. By incorporating a framework like LIME, the specialist would have not only a predicted diagnosis, but also the reason behind it for that specific prediction, meaning the features that played a major role when giving the output.

References

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²¹Post-hoc method that explains the individual predictions of complex (black-box) models by approximating their local behavior using an interpretable surrogate model (e.g., a linear model), trained on perturbed samples generated around the input of interest.

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